

Augustin's Reference Guide to Mathematical Set Theory in $\text{\LaTeX}$

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A comprehensive reference guide for formatting mathematical sets, operations, quantifiers, and intervals in $\text{\LaTeX}$. Curated and maintained by [Augustin Thomas](#).

All commands below must be wrapped in math mode delimiters ($\$$ or $\backslash[\]$).

Quick Reference

1. Common Number Sets (Blackboard Bold)

Note: To use these symbols, you must include `\usepackage{amssymb}` or `\usepackage{amsfonts}` in your document's preamble.

Set Name	$\text{\LaTeX}$ Code	Rendered Output
Natural Numbers	<code>\mathbb{N}</code>	$\mathbb{N}$
Integers	<code>\mathbb{Z}</code>	$\mathbb{Z}$
Rational Numbers	<code>\mathbb{Q}</code>	$\mathbb{Q}$
Real Numbers	<code>\mathbb{R}</code>	$\mathbb{R}$
Complex Numbers	<code>\mathbb{C}</code>	$\mathbb{C}$
Empty Set	<code>\emptyset</code> or <code>\varnothing</code>	$\emptyset$ or $\varnothing$

2. Core Set Operations

Operation	$\text{\LaTeX}$ Code	Rendered Output
Union	<code>A \cup B</code>	$A \cup B$
Intersection	<code>A \cap B</code>	$A \cap B$
Set Difference / Minus	<code>A \setminus B</code>	$A \setminus B$
Cartesian Product	<code>A \times B</code>	$A \times B$
Symmetric Difference	<code>A \triangle B</code>	$A \triangle B$
Complement	<code>A^c</code> or <code>A'</code>	A^c or A'

3. Relations and Membership

Meaning	L ^A T _E X Code	Rendered Output
Element of / Belongs to	<code>x \in A</code>	$x \in A$
Not an element of	<code>x \notin A</code>	$x \notin A$
Subset of (or equal to)	<code>A \subseteq B</code>	$A \subseteq B$
Strict / Proper Subset	<code>A \subset B</code> or <code>A \subsetneq B</code>	$A \subset B$ or $A \subsetneq B$
Superset of	<code>A \supseteq B</code>	$A \supseteq B$
Not a subset of	<code>A \not\subseteq B</code>	$A \not\subseteq B$

4. Quantifiers and Logic

Meaning	L ^A T _E X Code	Rendered Output
For all / For each	<code>\forall</code>	$\forall$
There exists	<code>\exists</code>	$\exists$
There exists a unique	<code>\exists!</code>	$\exists!$
Does not exist	<code>\nexists</code>	$\nexists$
Implies	<code>\implies</code>	$\implies$
If and only if (Equivalent)	<code>\iff</code>	$\iff$
Therefore	<code>\therefore</code>	$\therefore$
Because / Since	<code>\because</code>	$\because$

5. Intervals and Bounds

Interval Type	L ^A T _E X Code	Rendered Output
Closed Interval	<code>[a, b]</code>	$[a, b]$
Open Interval	<code>(a, b)</code>	(a, b)
Left-Open, Right-Closed	<code>(a, b]</code>	$(a, b]$
Left-Closed, Right-Open	<code>[a, b)</code>	$[a, b)$
Infinity	<code>\infty</code>	∞
Negative Infinity	<code>-\infty</code>	$-\infty$

6. Power Sets, Collections, and Families

When denoting families of sets, collections, topologies, or power sets, it is standard practice to use a calligraphic font. You can achieve this using the `\mathcal{}` command.

Concept	L ^A T _E X Code	Rendered Output
Power Set	<code>\mathcal{P}(A)</code> or <code>2^A</code>	$\mathcal{P}(A)$ or 2^A
Family of Sets	<code>\mathcal{F}</code>	$\mathcal{F}$
Collection of Sets	<code>\mathcal{C}</code>	$\mathcal{C}$
Topology	<code>\mathcal{T}</code>	$\mathcal{T}$

Large Indexed Operations

Indexed Union

- **Inline Mode** (`$...$`): `$\bigcup_{i=1}^n A_i$` $\rightarrow \bigcup_{i=1}^n A_i$
- **Display Mode** (`\[...\]`): `\[ \bigcup_{i=1}^n A_i \]` $\rightarrow$

$$\bigcup_{i=1}^n A_i$$

Indexed Intersection

- **Inline Mode** (`$...$`): `$\bigcap_{i=1}^n A_i$` $\rightarrow \bigcap_{i=1}^n A_i$
- **Display Mode** (`\[...\]`): `\[ \bigcap_{i=1}^n A_i \]` $\rightarrow$

$$\bigcap_{i=1}^n A_i$$

Set-Builder Notation & Braces

Because curly braces `{ }` are used for grouping code in $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$, you must escape them with a backslash `\{ \}` to display them visually.

- **Standard Notation:**

`\{ x \in \mathbb{R} \mid x > 0 \}`

Output: $\{x \in \mathbb{R} \mid x > 0\}$

- **Auto-Scaling Notation (For tall elements like fractions):**

`\left\{ x \in \mathbb{Q} \; ; \middlevert x = \frac{1}{n} \right\}`

Output: $\left\{x \in \mathbb{Q} \mid x = \frac{1}{n}\right\}$

Formatting Tips

1. Text Inside Math Mode

Use `\text{...}` (requires `\usepackage{amsmath}`) to add normal words inside your equations so they do not render in cramped math italics:

`\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} \quad \text{if } x < y`

Output: $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z} \quad \text{if } x < y$

2. Fine-Tuning Math Spacing

If your symbols look too crowded, inject small spaces using these spacer commands:

- `\,` (thin space)
- `\;` (medium space)
- `\quad` (large space equal to the width of one character)

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